

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Integrate each of the following.

a) $\int_{-\pi/2}^{\pi/2} \cos^7(x) dx$

b) $\int_{-\pi/2}^{\pi/2} \cos^7(x) \sin^3(x) dx$

c) $\int_{-\pi/2}^{\pi/2} \cos^4(x) dx$

d) $\int_0^{\pi/6} \sqrt{1 + \cos(2x)} dx$

2. Evaluate each of the following integrals.

a) $\int \tan(x) dx$. (No Calc 2 techniques are needed here)

b) $\int \tan^2(x) dx$

c) $\int \tan^3(x) dx$ (No Calc 2 techniques are needed here)

d) $\int \tan^3(x) \sec^3(x) dx$

3. Consider the integral $\int \frac{10x}{\sqrt{3+x^2}} dx$.

a) Evaluate this integral.

b) Would it have been okay to use the angle restriction of $\frac{3\pi}{2} < \theta < \frac{5\pi}{2}$? What, if anything, would this change about your solutions from part (a)?

c) Suppose that our antiderivative contained θ . Would it be possible to replace theta with $\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$?

d) Suppose that our antiderivative contained θ . Would it be possible to replace theta with $\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$ if we were using the angle restriction $\frac{3\pi}{2} < \theta < \frac{5\pi}{2}$?

4. Evaluate the integral $\int_1^3 \frac{10x}{\sqrt{3+x^2}} dx$.

5. Evaluate the definite integral $\int_0^2 x\sqrt{1-(x-1)^2} dx$.

6. Evaluate the integral $\int_2^4 \sqrt{(x-2)(4-x)} dx$.